Course Code: CSE3025	**Course Name:** Making Causal Inferences to Support Decisions	
Credits: 3 (2L-0T-2P)	**Contact Hours: 3** sessions per week	
Batch: 2023, 6th Sem **Academic Year:** 2026-27	**Semester Duration:** 19 Jan 2026 to 2 Jun 2026	
Course Faculty: Prof. Anantha Rao 35 E2 4th floor Mobile: 9868160550	**Course Coordinator:** Prof. Anantha Rao 35 E2 4th floor Mobile: 9868160550	

Aim of the course: Decision-makers when presented with results of machine learning models often ask why a specific outcome was observed. A common approach is to bring out the influence of features/ independent variables on the response variable and avoid invoking causality. This course aims to fill that gap. The course will present a broad framework within which casual inferences - using data and a firm's internal or institutional knowledge – can be made to support decisions. Students will learn to apply the concepts of causal inference to real world data, so that they can extend their data science toolkit. The course uses theoretical explanations along with practical examples implementable in Python code to give students a realistic feel.

Course Overview and Context: Making Causal Inferences is an elective course for Computer Science and Engineering streams. The course will introduce students to building blocks of causal inference framework for linear and non-linear relationships. It will focus on estimating causal effects from observational data in conjunction with inputs from firm's managers and their institutional knowledge in a systematic way.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand the concepts & building blocks to specify a causal inference model	Employability	→ theory sessions. → In-class tool demonstrations → Industry case studies etc. → Through Quiz and Assignments
CO2	Develop structural causal models to represent decision problems	Skill Development	→ Hands-on practical sessions in labs. → Developing Graphical Models via magjac and graphviz library. → Through Quiz and Assignments
CO3	Analyze models by testing for assumptions and convergence with causal knowledge	Entrepreneurship, Skill Development	→ Modelling with structural and graphical approaches on datasets and implement via matplotlib , statsmodels python libraries. → Through Quiz and Assignments

CO4	Apply causal models by analyzing decision situations to across different verticals	Entrepreneurship, Skill Development	→ Comparing models on industry specific datasets using advance python libraries (DoWhy, EconML, gCastle). → Through End Semester Project.

Prerequisite: Probability, Statistics, Machine Learning, Python Programming

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Introduction to Causal inference paradigm	**6(3 Th + 3 Lab)**	CO1
Concepts of Ladder of Causation	**6(3 Th + 3 Lab)**	CO1
Classical Regression and Statistical Control and its limitations	**6(3 Th + 3 Lab)**	CO1, CO2
Structural Causal Models: Forks, Chains & Colliders; and Conditional Independence	**6(3 Th + 3 Lab)**	CO2
Causal Inference Process, Assumptions and Challenges	**6(3 Th + 3 Lab)**	CO2, CO3
Causal Inference in non-linear models: Estimators, Experiments and Evaluation of Models	**6(3 Th + 3 Lab)**	CO3, CO4
Causal Discovery – Applications	**6(3 Th + 3 Lab)**	CO3, CO4

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	1			1								2		3	
CO2	3	2	3		3	2	2						3		2	
CO3	3	2	3		3								3		2	
CO4		3			2								3		2	

Learning Resources:

Textbooks:

1. Aleksander Molak, ***"Causal Inference and Discovery in Python",*** Packt.
2. Judea Pearl, Madelyn Glymour & Nicholas Jewell "***Causal Inference in Statistics: A Primer ", Wiley.***

Reference Books:

3. Judea Pearl & Dana Mackenzie, ***The Book of Why***. Penguin Science.
4. Hernan, MA & Robins, JM. ***Causal Inference What If***. CRC.
5. Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. ***Elements of causal inference: foundations and learning algorithms***. The MIT Press.

Video lectures/ MOOC:

1. UCLA Office of Advanced Research Computing - ***What is Causal Inference and Where is Data Science Going*** https://www.youtube.com/watch?v=MNyI1Xkapxg&t=425sh
2. Judea Pearl, ***New Science of Cause and Effect or Seven Pillars of Causal Wisdom*** https://www.youtube.com/watch?v=ZaPV1OSEpHw

Reference Links:

1. Brian Scholkpf ***Causality for Machine Learning*** https://dl.acm.org/doi/10.1145/3501714.3501755
2. http://magjac.com/graphviz-visual-editor/; online graphviz interface

Note: Additional resources, wherever required will be posted on classroom.

Experiential Learning Component:

The project, as mentioned in the above table, is the primary experiential learning component of this course. A student needs to spend approximately 15 to 20 hours to finish this assignment. The weightage of this component is 40%. The programming assignments (30% weightage) will help develop problem formulating and solving abilities. Overall, the major focus of this course is on hands-on experiential learning. The total weightage to experiential learning is 70%.

Assessment Pattern: The final grade will be based on the marks / grades obtained in the quiz, assignments, project along with other assessments defined in the assessment table below.

A relative grading method defined in the academic regulations of the university will be followed to grade the students. Students must secure a minimum of 40% of marks after completing all the assessments in the following table to become eligible for grading.

	Component	Duration (mins)	Weightage (%)	Evaluation Week/Pattern
Quiz	Quiz 1, 2	20 mins each	2 x 15% = 30%	Throughout the semester. The date of the quiz will be announced one week in advance.
Lab Evaluation	Programming Assignments 1, 2	Scheduled Lab hours	2 x 15% = 30%	Throughout the semester. Evaluation will be based on the programming assignments provided during the LAB sessions.
Project	End-Term Project Evaluation	Take Home	1x 40% = 40%	Project report will be required along with Colab NB solutions, followed by presentation / viva. Rubrics: 10 + 20 +10

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.